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office: A5304

introduction to probability by Bertsekas and Tsitsiklis

surprise quiz 10%

! NCERT prerequisites

Random experiment = experiment involving randomness

Sample space = set of all possible outcomes of an RE (denoted by Ω)

$$\Omega_c = \{H, T\} \quad \text{coin toss}$$

$$\Omega_{2c} = \{H, T\} \times \{H, T\} \quad \text{tossing two coins}$$

Sample point / outcome = an element $\omega \in \Omega$ Event = a subset $A \subseteq \Omega$ $\mathbb{P}(A)$ = probability of event A

SET FUNCTION

zero probability $\neq$ impossiblee.g. choosing a number in $[0, 1]$

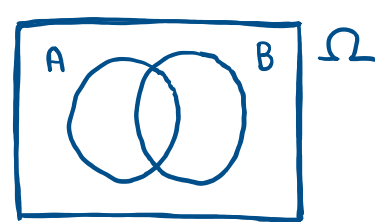
$$\mathbb{P}(\{0.5\}) = 0$$

but not impossible

$$A \setminus B = A - A \cap B = A \cap B'$$
 set difference

$$\text{symmetric difference} = (A \setminus B) \cup (B \setminus A)$$

$$\text{mutually exclusive: } A \cap B = \emptyset$$

cardinality of $A = |A|$

$$\text{inclusion-exclusion principle: } |A \cup B| = |A| + |B| - |A \cap B|$$

$$\text{FOR } N \text{ SETS: } \left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n-1} \left| \bigcap_{i=1}^n A_i \right|$$

• countable sets = finite

= 1-1 correspondence with natural numbers

• MONOTONE SEQUENCE OF SETS

increasing sequence: $A_1 \subseteq A_2 \subseteq A_3 \dots$ decreasing sequence: $A_1 \supseteq A_2 \supseteq A_3 \dots$

$$I_n = \left[0, 1 - \frac{1}{n}\right] \quad D_n = \left[0, \frac{1}{n}\right]$$

Powerset of $A = \mathcal{P}(A)$ = set of all subsets of A

$$|A| = n$$

$$|\mathcal{P}(A)| = 2^n$$

$$\left\{ \begin{array}{l} \text{probability } \mathbb{P} \\ \text{power set } \mathcal{P} \end{array} \right\}$$

$$\mathcal{P}([0, 1]) = \left\{ \begin{array}{l} (x, y) \\ [x, y] \\ [x, y) \\ (x, y] \end{array} : 0 < x < y < 1 \right\}$$

functions = mappings from domain to range

$$f: \mathcal{D} \rightarrow \mathcal{R}$$

set function $\Rightarrow$ domain is a set of sets

FOR PROBABILITY FNS, THE POWER SET OF THE SAMPLE SPACE CAN BE DOMAIN.

e.g. length of closed segments on the real line.

PROBABILITY AXIOMS

$$\textcircled{1} \quad \mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\Omega) = 1$$

$$\textcircled{2} \quad \text{For a set } A \subseteq \Omega, \quad 0 \leq \mathbb{P}(A) \leq 1$$

$$\textcircled{3} \quad \text{For a disjoint collection of events } A_1, A_2, \dots \quad (A_i \subseteq \Omega)$$

countable

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

countable union

cannot apply to cases like choosing a number in $[0, 1]$ Probability measure $\mathbb{P}$ is a set function $\mathbb{P}: \mathcal{P}(A) \rightarrow [0, 1]$

that satisfies the above axioms.

• non-measurable sets?